Remark

There are functions for which Rolle's theorem may not be applicable.

(1) For the function $f(x)=|x|, x\in[-1,1]$ Rolle's theorem is not applicable, even though $f(-1)=1=f(1)$ because $f(x)$ is not differentiable at $x=0$.

(2) For the function,

$$f(x)=\begin{cases}1 & \text{when } x=0,\\ x & \text{when } 0<x\le 1\end{cases}.$$

even though $f(0)=f(1)=1$, Rolle's theorem is not applicable because the function $f(x)$ is not continuous at $x=0$.

(3) For the function $f(x)=\sin x,\ x\in\left[0,\frac{\pi}{2}\right]$ Rolle's theorem is not applicable, even though $f(x)$ is continuous in the closed interval $\left[0,\frac{\pi}{2}\right]$ and differentiable in the open interval $\left(0,\frac{\pi}{2}\right)$ because, $0=f(0)\neq f\left(\frac{\pi}{2}\right)=1$.

If $f(x)$ is continuous in the closed interval $[a,b]$ and differentiable in the open interval (a,b) and even if $f(a)\neq f(b)$ then the Rolle's theorem can be generalised as follows.

7.3.2 Lagrange's Mean Value Theorem

Theorem 7.3

Let $f(x)$ be continuous in a closed interval $[a,b]$ and differentiable in the open interval (a,b) (where $f(a), f(b)$ are not necessarily equal). Then there exist at least one point $c\in(a,b)$ such that,

$$f'(c) = \frac{f(b)-f(a)}{b-a} \qquad \ldots (6)$$

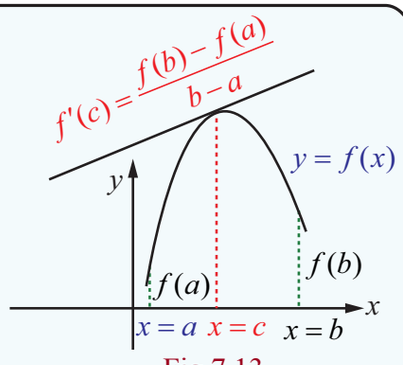


Fig.7.13

Remark

If $f(a)=f(b)$ then Lagrange's Mean Value Theorem gives the Rolle's theorem. It is also known as **rotated Rolle's Theorem**.

Remark

A physical meaning of the above theorem is the number $\frac{f(b)-f(a)}{b-a}$ can be thought of as the average rate of change in $f(x)$ over (a,b) and $f'(c)$ as an instantaneous change.

A geometrical meaning of the Lagrange's mean value theorem is that the instantaneous rate of change at some interior point is equal to the average rate of change over the entire interval. This is illustrated as follows :

Chapter 7 Differential Calculus Original-new.indd 18 26-02-2020 17:22:53